

**HUANG, Chun-Jen IN THE UNITED STATES PATENT AND TRADEMARK OFFICE**

|                                                             |                           |
|-------------------------------------------------------------|---------------------------|
| First Named Inventor: HUANG, Chun-Jen                       | Examiner:                 |
| Application No.: 19/060,464                                 | Art Unit:                 |
| Filed: Feb 21, 2025                                         | Confirmation Number: 9608 |
| Title: COMPOSITION AND METHOD FOR<br>TREATING LUNG DISEASES |                           |
| Attorney Docket No.: 5845/0104PUS2                          |                           |

**DECLARATION UNDER 37 C.F.R. § 1.132**

I, **HUANG, Chun-Jen**, do hereby declare as follows:

1. I am a [ professor at the graduate institute of clinical medicine, Taipei Medical University, specializing in biomedical research with a focus on [e.g., regenerative medicine, exosome-based therapeutics, and translational pharmacology.]
2. [I received my Ph.D. in July 2004 from the Graduate Institute of Pharmacology at National Yang-Ming University, Taipei, Taiwan.]
3. [Since August 2017, I have served as an Attending Anesthesiologist at the Department of Anesthesiology, Wan Fang Hospital, and since October 2017, as a Professor at the Graduate Institute of Clinical Medicine, Taipei Medical University. From 2009 to 2017, I worked at Taipei Tzu Chi Hospital as an Attending Physician and Department Head. In addition, from April 2012 to July 2017, I served as a Professor at the School of Medicine, Tzu Chi University, Hualien, Taiwan.]
4. I have been asked by the inventor of the above-identified application to give my opinion of the properties of engineered exosomes loaded with hsa-let-7i-5p microRNA as disclosed in the present application. In particular I have been asked to evaluate their characterization, biodistribution, pharmacokinetics, and therapeutic effects in preclinical models of lung injury. I have been compensated for my time in performing this review at my usual rate plus expenses.
5. I performed the following tests. [Engineered exosomes were prepared from murine RAW264.7 macrophage cells genetically modified to overexpress hsa-let-7i-5p microRNA using plasmid transfection and electroporation. Exosomes were isolated via

differential ultracentrifugation and characterized using transmission electron microscopy for morphology, nanoparticle tracking analysis for size distribution, and Western blotting (Simple Western method) for specific surface markers such as CD63, CD9, and ALIX. Quantification of hsa-let-7i-5p miRNA levels was conducted using next-generation sequencing (NGS) and droplet digital PCR (ddPCR). For functional assessment, C57BL/6 mice were subjected to lipopolysaccharide (LPS)-induced sepsis or aspiration pneumonia to model acute lung injury. Engineered exosomes were administered intraperitoneally or intratracheally. Therapeutic efficacy was evaluated by measuring survival rates, lung histopathology (H&E staining and lung injury scoring), pulmonary edema (wet/dry weight ratio), lung function (via FlexiVent system), and inflammatory cell infiltration (via BALF analysis).]

6. I examined the results. [The engineered exosomes exhibited consistent morphology, particle size (approximately 100 nm), and expression of exosomal markers. High levels of hsa-let-7i-5p were detected in engineered exosomes compared to unmodified controls. Biodistribution assays using Cy7-labeled exosomes revealed significant accumulation in lung tissues following both intraperitoneal and intratracheal administration, with an in vivo half-life extending up to 48 hours. In murine models of sepsis and aspiration pneumonia, administration of engineered exosomes significantly improved 48-hour survival rates, reduced histological lung damage, lowered pulmonary edema, and improved lung function parameters. BALF analysis demonstrated reduced total cell counts and inflammatory markers, indicating attenuation of lung inflammation.]
7. [Based on the above findings, I conclude that the engineered exosomes loaded with hsa-let-7i-5p microRNA possess favorable physicochemical properties, exhibit targeted biodistribution to lung tissues, and demonstrate robust therapeutic efficacy in preclinical models of acute lung injury. These findings support the claims made in the present application regarding their potential use in the prevention and treatment of acute respiratory distress syndrome (ARDS) and related pulmonary conditions.]
8. I hereby declare that all statements made herein of my own knowledge are true and that all statements based on information and belief are believed to be true and further that these statements have been made with the knowledge that willful false statements and the like so made are punishable by fine, imprisonment, or both under section 1001 of Title 18

of the United States Code and that such false statements may jeopardize the validity of the application or any patent issuing thereon.

HUANG, Chun-Jen

Name

Ch-Jen Hg

Signature

2025/04/18

Date

**IN THE UNITED STATES PATENT AND TRADEMARK OFFICE**

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**DECLARATION UNDER 37 C.F.R. § 1.132**

I, **CHANG, Chao-Yuan**, do hereby declare as follows:

1. I am a [Medical Research Scientist affiliated with the Department of Research at Wan Fang Hospital].
2. [I received my Ph.D. from the Graduate Institute of Biotechnology, National Chung Hsing University, Taichung, Taiwan, in June 2013.]
3. [I have been employed as a Medical Researcher at the Department of Research, Wan Fang Hospital, since August 2021. From 2017 to 2021, I worked as a Postdoctoral Research Fellow at the Graduate Institute of Clinical Medicine, Taipei Medical University, Taipei, Taiwan.]
4. I have been asked by the inventor of the above-identified application to give my opinion of the properties of engineered exosomes loaded with hsa-let-7i-5p microRNA as disclosed in the present application. In particular I have been asked to evaluate their characterization, biodistribution, pharmacokinetics, and therapeutic effects in preclinical models of lung injury. I have been compensated for my time in performing this review at my usual rate plus expenses.
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Quantification of hsa-let-7i-5p miRNA levels was conducted using next-generation sequencing (NGS) and droplet digital PCR (ddPCR). For functional assessment, C57BL/6 mice were subjected to lipopolysaccharide (LPS)-induced sepsis or aspiration pneumonia to model acute lung injury. Engineered exosomes were administered intraperitoneally or intratracheally. Therapeutic efficacy was evaluated by measuring survival rates, lung histopathology (H&E staining and lung injury scoring), pulmonary edema (wet/dry weight ratio), lung function (via FlexiVent system), and inflammatory cell infiltration (via BALF analysis).]

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CHANG, Chao-Yuan

Name

Chao Y. Chang

Signature

2025/04/18

Date